

Assignment 1: Generalized linear models

ADVANCED STATISTICAL LEARNING AND MODELING - MODULE B / Giulia Vannucci

2025-10-27

Instructions

This assignment consists of three exercises involving the application of different type of GLMs covered during the course. Students are free to use either R or Python to complete the assignments.

Assignments may be completed individually or in groups. However, each student must submit their own version of the assignment by uploading it to the following directory on Microsoft Teams:

Assignments → **Assignment1** → [create a folder named: `name_surname_assignment1`]

Each student's folder must contain:

- A PDF report including both exercises;
- The code used for the analysis (in '.R', '.py', or '.ipynb' format);
- A note stating whether the work was done individually or as part of a group, and if in a group, the names of all group members.

Deadline: the day of the end of exam inscription. While there is no penalty for late submission, students are strongly encouraged to complete the assignment as early as possible to allow time for feedback and clarification.

Exercise 1: Modeling Student Performance with Categorical Logistic Models

Dataset: UCI Student Performance Data Set

This assignment focuses on the student performance data (use the `student-mat.csv` file, concerning the Math course, for simplicity). The target variable is the final grade, $G3$.

Objective

The goal is to model the final student performance ($G3$ categorized) using key social, behavioral, and demographic predictors. Students must define a categorical response variable and then apply both **Multinomial** and **Ordinal Logistic Regression** models to analyze and compare the results.

Instructions

1. Data Preparation and Categorical Target Definition

- Load and inspect the `student-mat.csv` dataset.
- **Categorization of $G3$:** The final grade ($G3$, ranging from 0 to 20) must be converted into a new **categorical outcome variable** with at least **three levels** (e.g., "Low Performance," "Medium Performance," "High Performance").

- **Decision Point:** Clearly justify the choice of categories and the corresponding grade thresholds (e.g., why thresholds at 10 and 15?).
 - Select and prepare the predictors: choose predictors and ensure all predictors are correctly encoded as factors or appropriate numerical types.
2. **Modeling: Multinomial vs. Ordinal**
 - **Multinomial Logistic Regression:** Fit a multinomial logistic regression model to predict the categorized *G3* outcome (treating the categories as **nominal**).
 - **Ordinal Logistic Regression:** Fit an ordinal logistic regression model (Proportional Odds Model) to predict the same categorized *G3* outcome (treating the categories as **ordinal**).
 3. **Model Comparison and Interpretation**
 - **Proportional Odds Assumption:** For the ordinal model, state and **test the assumption of Proportional Odds**.
 - **Coefficient Interpretation:** Interpret the coefficients in both the Multinomial and Ordinal models.
 - **Model Choice:** Use appropriate statistical criteria (e.g., Likelihood Ratio Test, AIC, BIC) to compare the multinomial and ordinal models.
 4. **Report**
 - Detail the process and justification for the categorization of the *G3* variable.
 - Discuss the statistical results of the **Proportional Odds test** and its implication for the interpretability of the Ordinal model.
 - Interpret the key coefficients for the Ordinal Model, explaining what the proportional odds assumption implies for the effect of a predictor across different performance splits.
 - Justify your final choice between the Multinomial and Ordinal models for describing student performance.

Exercise 2: Log-Linear Models and Equivalence with Logistic Regression

Dataset: UCI Adults Dataset (Focus on categorical variables only)

This dataset contains demographic information. For this exercise, focus only on the categorical variables.

Objective

The goal is to study the associations among key demographic factors using **Log-Linear Models** and formally demonstrate their mathematical relationship to a **Binary Logistic Regression** model.

Instructions

1. **Data Preparation**
 - Load the dataset and select three key categorical variables: **Sex** (Gender), **Race**, and **Education**. Recode **Education** into fewer, more manageable categories (e.g., 'HighSchool', 'College', 'Advanced').
 - Create and inspect the **Contingency Table** cross-classifying the three selected variables.
2. **Log-Linear Modeling**
 - Fit and compare three nested log-linear models (e.g., the mutually independent model, the conditional independence model, and the model containing all two-way associations).
 - Use the **Likelihood Ratio Chi-Square** statistic to perform formal hypothesis tests for model comparison and assess overall model fit.
 - Interpret the parameters of the most parsimonious acceptable model, focusing on any significant two-way interaction terms.
3. **Equivalence Demonstration**
 - Fit a **Binary Logistic Regression** model predicting **Sex** as a function of **Race** and the recoded **Education** (only including main effects).

- Identify the specific log-linear model structure that is theoretically equivalent to the fitted logistic model.
 - Show how the estimated log-linear parameters from the equivalent log-linear model relate mathematically (in terms of magnitude and sign) to the estimated logistic regression coefficients.
4. **Report**
- Discuss the conditions under which a log-linear model for a three-way table is mathematically equivalent to a logistic regression model.
 - Provide a comparison table of the estimated coefficients (and standard errors, if feasible) from the two equivalent models to demonstrate the mathematical link.

Exercise 3: Count Data Modeling with Overdispersion and Zero-Inflation

Dataset: UCI Absenteeism at work

The target variable is the number of **Absenteeism Time in Hours**. This count data often exhibits an excess of zeros.

Objective

The goal is to model the number of absent hours using standard count models (Poisson, Negative Binomial) and to specifically address the issue of **Zero Inflation** using an appropriate two-part model.

Instructions

1. **Data Exploration and Diagnostics**
 - Load and inspect the dataset. Use **Absenteeism Time in Hours** as the count response variable (Y).
 - Select relevant predictors (e.g., **Distance to Work**, **Age**, **Service Time**, **Education**, **Reason for absence** categories).
 - Calculate the mean and variance of Y and formally check for **overdispersion** and **zero inflation**.
2. **Model Fitting and Selection**
 - Fit a **Poisson Regression Model**.
 - Fit a **Negative Binomial Regression Model** (to handle general overdispersion).
 - Fit a **Zero-Inflated Poisson (ZIP)** or **Zero-Inflated Negative Binomial (ZINB)** model.
3. **Model Diagnostics and Interpretation**
 - Use appropriate model comparison criteria (AIC, BIC) to select the best model among the alternatives.
 - For the final chosen zero-inflated model, interpret the two distinct sets of coefficients:
 - The coefficients of the **Count Component** (modeling the expected count given the employee is absent).
 - The coefficients of the **Zero-Inflation Component** (modeling the probability of being in the “always-zero” group).
4. **Report**
 - Justify the final model choice based on diagnostic checks (overdispersion, zero count) and fit statistics.
 - Interpret the two parts of the final model in a clear, practical context (e.g., factors affecting *if* an employee is never absent versus factors affecting *how many* hours they are absent).
 - Discuss the potential advantages of the zero-inflated model over the Negative Binomial model for this specific type of data.

Submission Checklist

In your personal folder on Teams, include:

- `assignment1_report.pdf` (including all three exercises);
- The script or notebook used (`.R`, `.py`, or `.ipynb`);
- A note specifying individual or group work and, if in a group, the names of all members.

Folder structure:

`Assignments` → `Assignment1` → `name_surname_assignment1`

Good luck, and don't hesitate to ask questions as you work through the assignments!